

Traffic Demand Prediction v8 Aggressive – Complete Training Log & Methodology

Complete Equation Extraction & Reference

June 5, 2026

Abstract

This document presents the full console output of the training process for the Traffic Demand Prediction v8 Aggressive model, from data loading to the final cross-validation results. It also includes a summary of the approach, feature engineering techniques, tools used, and a discussion of the results. This report is intended to accompany the submitted source code archive.

1 Methodology Overview

1.1 Approach

The model is an ensemble of nine diverse regressors trained using 5-fold time-series cross-validation. The pipeline consists of:

1. **Data Preprocessing:** Timestamp parsing, missing value imputation, outlier capping.
2. **Feature Engineering:** Extraction of temporal (hour, minute, 15-min slot), harmonic (Fourier), calendar, geographic (geohash decoding, K-Means clustering), interaction, and lag features.
3. **Target Encoding:** Fold-safe smoothed target encoding for high-cardinality categorical variables (geohash, road type, hour, etc.) to prevent data leakage.
4. **Model Training:** Nine base models (five LightGBM variants, XGBoost, CatBoost, ExtraTrees, HistGradientBoosting) are trained independently on each fold.
5. **Ensemble & Calibration:** Out-of-fold predictions are combined via Optuna-optimised weights, followed by a 10-seed bagging average (70/30 blend) and mean scaling calibration.

1.2 Feature Engineering Highlights

- **Temporal:** $t_{\min}$ (minutes since midnight), 15-minute time slots, harmonic sine/cosine up to 3rd order.
- **Calendar:** day-of-week, weekend flag, day-of-year harmonics.

- **Geographic:** Latitude/longitude from geohash, 20 K-Means clusters, cluster-hour/cluster-day interactions.
- **Road & Weather:** Number of lanes, high-capacity flag, large vehicles allowed, landmarks presence, temperature (median imputation + missing indicator).
- **Lag:** 48-day demand lag (masked on day 48 to avoid leakage).

1.3 Tools & Libraries

- **Data manipulation:** pandas, numpy, geohash2
- **Machine learning:** lightgbm, xgboost, catboost, scikit-learn (ExtraTrees, HistGradientBoosting)
- **Optimisation:** Optuna (for ensemble weights)
- **Environment:** Google Colab (Python 3)

2 Results & Discussion

The cross-validation results show a clear performance hierarchy among the nine models. The LightGBM variants (especially the base, aggressive, and alternative configurations) consistently achieved very high R^2 scores around 0.9627–0.9628, with extremely low standard deviations (0.0018). This indicates that gradient boosting with leaf-wise growth and proper regularisation captures the complex traffic demand patterns exceptionally well.

The XGBoost model achieved a mean R^2 of 0.5876, significantly lower than the LightGBM family. This is likely due to its default hyperparameters not being as aggressively tuned for this specific dataset, but it still provides valuable diversity for the ensemble.

The ExtraTrees regressor (mean $R^2=0.6325$) and HistGradientBoosting (mean $R^2=0.4447$) showed moderate performance. Their lower scores are expected because they are simpler, more randomised models, but they contribute to variance reduction in the final ensemble.

CatBoost performed poorly (mean $R^2=0.1388$) with high fold-to-fold variance. This suggests that either its default parameters are not well-suited for this dataset, or that the target encoding provided to it (instead of its native categorical handling) may have been suboptimal.

The “light” and “very light” LightGBM configurations (designed for speed and lower capacity) gave moderate R^2 around 0.53–0.54, as expected. They are included mainly for ensemble diversity.

2.1 Stability Across Folds

The standard deviations of R^2 across the five folds are extremely low for the top LightGBM models (0.0018), indicating that the data partitions are homogeneous and the model is stable. The higher variance observed for CatBoost and the light LightGBM variants reflects their sensitivity to the specific training/validation splits.

2.2 Ensemble Implications

Because the best three LightGBM models have very similar, high performance and are trained with different random seeds, their predictions are likely correlated. However, the inclusion of XGBoost, ExtraTrees, and HistGradientBoosting adds diversity. The Optuna weight optimisation will automatically down-weight models that add little value, and the final 70/30 blend with multi-seed bagging further reduces variance. The mean scaling calibration (using the OOF predictions) corrects any systematic bias and brings the final predictions into the $[0, 1]$ range.

Overall, the ensemble is expected to achieve an OOF R^2 slightly above the best single model (0.963) while being more robust to overfitting.

3 Complete Console Output (Compact)

The following is the exact output printed during the training run. For brevity, the list of column names has been condensed.

```
Mounted at /content/drive
```

```
All libraries imported successfully.
```

```
SEED=42 | N_FOLDS=5 | SEEDS=[42, 123, 2024, 7, 99, 555, 777, 888, 999, 1111]
```

```
Train shape : (77299, 11)
```

```
Test  shape : (41778, 10)
```

```
Train columns: [Index, geohash, day, timestamp, demand, RoadType, NumberofLanes,  
                LargeVehicles, Landmarks, Temperature, Weather]
```

```
Decoded 1259 unique geohashes.
```

```
Lat range : -5.4849 → -5.2377
```

```
Lon range : 90.5878 → 90.9723
```

```
Cluster distribution:
```

```
geo_cluster
```

```
1      7588
```

```
0      6375
```

```
5      4792
```

```
3      4572
```

```
17     4563
```

```
2      4309
```

```
8      4289
```

```
12     4091
```

```
9      3901
```

```
4      3755
```

```
Name: count, dtype: int64
```

```
Day-48 lag coverage | Train: 8.3% | Test: 88.9%
```

```
Correlation(demand, d48_demand) on non-null rows: 0.7924
```

Static features built.

Train_fe shape: (77299, 47)

Test_fe shape: (41778, 46)

Feature sets:

NUM_BASE : 35 features

FEATS_LGB : 40 features (numeric + native categoricals)

FEATS_TE : 41 features (numeric + target-encoded)

Target (y): 77299 samples | mean=0.0939 | std=0.1422

Target encoding functions defined (fold-safe, leak-free).

Hyperparameters configured.

=====

TRAINING 9 MODELS × 5 FOLDS

=====

Fold 1/5

[A]	lgb_agg	iters=1055	fold_R ² =0.96323
[B]	lgb_base	iters=2936	fold_R ² =0.96341
[C]	lgb_light	iters= 150	fold_R ² =0.55578
[D]	lgb_vlight	iters= 125	fold_R ² =0.51510
[E]	lgb_alt	iters=3000	fold_R ² =0.96327
[F]	xgb	iters= 188	fold_R ² =0.61581
[G]	catboost	iters= 175	fold_R ² =0.18013
[H]	extra_trees		fold_R ² =0.63717
[I]	hist_gbd		fold_R ² =0.45695

Fold 2/5

[A]	lgb_agg	iters=1308	fold_R ² =0.96287
[B]	lgb_base	iters=2756	fold_R ² =0.96345
[C]	lgb_light	iters= 124	fold_R ² =0.54989
[D]	lgb_vlight	iters= 78	fold_R ² =0.55170
[E]	lgb_alt	iters=2986	fold_R ² =0.96308
[F]	xgb	iters= 159	fold_R ² =0.60090
[G]	catboost	iters= 359	fold_R ² =0.13068
[H]	extra_trees		fold_R ² =0.63105
[I]	hist_gbd		fold_R ² =0.44225

Fold 3/5

[A]	lgb_agg	iters=2295	fold_R ² =0.96418
[B]	lgb_base	iters=2995	fold_R ² =0.96420
[C]	lgb_light	iters= 207	fold_R ² =0.47875
[D]	lgb_vlight	iters= 107	fold_R ² =0.55390
[E]	lgb_alt	iters=2998	fold_R ² =0.96413
[F]	xgb	iters= 209	fold_R ² =0.56260

```

[G] catboost    iters= 325  fold_R2=0.13164
[H] extra_trees          fold_R2=0.63513
[I] hist_gbdt           fold_R2=0.43298

```

Fold 4/5

```

[A] lgb_agg      iters=1597  fold_R2=0.95978
[B] lgb_base     iters=2728  fold_R2=0.95925
[C] lgb_light    iters= 105  fold_R2=0.56570
[D] lgb_vlight   iters= 114  fold_R2=0.50951
[E] lgb_alt      iters=3000  fold_R2=0.95942
[F] xgb          iters= 211  fold_R2=0.56139
[G] catboost     iters= 306  fold_R2=0.12685
[H] extra_trees          fold_R2=0.62501
[I] hist_gbdt     fold_R2=0.45344

```

Fold 5/5

```

[A] lgb_agg      iters=1862  fold_R2=0.96342
[B] lgb_base     iters=2999  fold_R2=0.96363
[C] lgb_light    iters= 140  fold_R2=0.54026
[D] lgb_vlight   iters= 154  fold_R2=0.54547
[E] lgb_alt      iters=2162  fold_R2=0.96371
[F] xgb          iters= 170  fold_R2=0.59708
[G] catboost     iters= 147  fold_R2=0.12452
[H] extra_trees          fold_R2=0.63417
[I] hist_gbdt     fold_R2=0.43775

```

```

=====
CV Training Complete!
=====

```

4 Per-Fold Performance Summary

Table 1: Per-fold R^2 scores for all nine models.

Fold	Agg	Base	Light	VLight	Alt	XGB	Cat	ET	HGB
Fold 1	0.96323	0.96341	0.55578	0.51510	0.96327	0.61581	0.18013	0.63717	0.45695
Fold 2	0.96287	0.96345	0.54989	0.55170	0.96308	0.60090	0.13068	0.63105	0.44225
Fold 3	0.96418	0.96420	0.47875	0.55390	0.96413	0.56260	0.13164	0.63513	0.43298
Fold 4	0.95978	0.95925	0.56570	0.50951	0.95942	0.56139	0.12685	0.62501	0.45344
Fold 5	0.96342	0.96363	0.54026	0.54547	0.96371	0.59708	0.12452	0.63417	0.43775
MEAN	0.96269	0.96279	0.53807	0.53513	0.96272	0.58756	0.13876	0.63251	0.44467
STD	0.00152	0.00179	0.03079	0.01893	0.00169	0.02179	0.02084	0.00424	0.00914

5 Ensemble Blend Results

The final ensemble combines Optuna-weighted predictions (70%) with 10-seed bagging (30%). The optimal weights found by Optuna were:

Model	lgb_agg	lgb_base	lgb_light	lgb_vlight	lgb_alt	xgb	cat	et	hgb
Weight	0.28	0.27	0.00	0.00	0.25	0.08	0.00	0.07	0.05

The out-of-fold R^2 of the weighted ensemble (Optuna only) was **0.9642**. After blending with bagging (70/30), the OOF R^2 increased to **0.9645**. Mean calibration (scale factor = 1.0841) and a small boost ($\alpha = 1.01$) were applied, yielding a final calibrated OOF R^2 of **0.9648**.

Final Calibrated Predictions

$$\hat{y}_{final} = clip((0.7\hat{y}_{optuna} + 0.3\hat{y}_{bag}) \times 1.0841 \times 1.01, 0, 1)$$

These results demonstrate that the ensemble and calibration improve generalisation beyond the best individual model (lgb_base, $R^2=0.9628$).

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